RYAN PEACH

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CONTACT

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PROFILE

OBJECTIVE: Relocating to continue a career in Computer Science and Electrical Engineering while pursuing my Masters, eventually pursuing a PHD in Cognitive Computing.
AVAILABILITY: Beginning Summer 2016

EDUCATION & AWARDS

Masters of Science in Computer Science - Robotics & Perception / Interactive Intel. *December 2017*
Georgia Tech. Institute of Technology, Atlanta, GA
Bachelor of Science in Electrical Engineering, Mathematics Minor *December 2014*
Western Kentucky University, Bowling Green, KY - GPA: 3.4

WORK EXPERIENCE

Nasco-NHK, Bowling Green, KY *May 2015 - Present*
Electrical Engineer (Controls & Automation)
- Programmed PLC's and robotics for a variety of control & automation applications.
- Designed cabinets using AutoCAD, implemented hardware upgrades to equipment.
- Diagnosed and maintained industrial electrical equipment.

Institute for Combustion Science and Technology, WKU *January 2015 – May 2015*
Electrical Engineer (Signals & Amplification)
- Worked with physicists and other researchers in designing rigs for experimentation.
- Helped design applications of Atomic Force Microscopes, Piezoelectric material, Signal Amplifiers, and Terra-hertz Lasers. Designed PCB's & programmed FPGA's.

Applied Physics Institute, Center for Research & Development, WKU *May 2014 – December 2014*
Student Electrical Engineer (PCB Design, Embedded Systems)
- Designed PCB's, performed research for both scientific and consumer purposes.
- Implemented chemical sensor arrays and machine learning algorithms.

Department of Chemistry, Western Kentucky University *January - March, 2014*
Student Software Developer / Electrical Engineer
- Designed software to control local multimeter via user friendly LabVIEW computer interface.
- Created applications for graphing Voltage, Resistance, & Current Density in real-time.
- Designed testing rig and product enclosure for solar array.

DHS-STEM Student Internship, **Pacific Northwest National Laboratory** *June - August, 2013*
Summer Intern (Social Data Analyst)
- Ran discrete-time Fourier analyses, digital filtering algorithms, and statistical analyses on large time-series datasets in Python and MatLab.
- Implemented machine learning spectral algorithms on large NLP datasets.

Lego Robotics Camp, Center for Gifted Studies, ***WKU*** *June, 2012; June, 2014*
Robotics Teacher (Lego Mindstorms, NXT Programming)

KEY SKILLS

Java, Scala, Python (Numpy, SciPy, Pandas, etc.), C/C++, Assembly, PLC Ladder Logic, Structured Ladder, Structured Text, SFC, LabVIEW (CLAD), MatLab, Simulink, Git, LaTeX, Linux, Android, HTML/CSS, VHDL, Verilog, OpenSCAD, 3D Printers

RS232/485, Modbus, Arduino, Allen-Bradley, Mitsubishi, Motoman, Proface HMI's, AutoCAD, MultiSIM, Eagle, Maxwell 3D, Unix, Linux, Windows, MS Office, Electronics / PCB Design, Microprocessors, AC/DC Motor Control, Cognitive Computing, SLAM,

Robotics, Computer Vision, Artificial Intelligence, Machine Learning, ASCIC FPGA Design, Physics & Mathematics, Computer Programming, Control Systems, Signal Analysis, High Quality Signal Amplification, Digital Signal Acquisition, Networking

SCHOOL & PROFESSIONAL PROJECTS

SCADA Senior Design Project (*WKU*) *Fall 2013 - Spring 2014*

- Created three PLC testbed systems: a traffic light, a water pump, and a power grid.
- Remotely controlled each system via SCADA.
- Performed simulated attacks on each system, and creating contingency programming to both prevent and react to each attack.

Motor Drives Control Project (*WKU*) *Spring 2013*

- Contributed to the writing of nine LabVIEW scripts to control and measure the speed, power, torque, voltage, and current supplied to/measured by a motor drive system.
- Created host-controlled VIs for top-down processing, and user interfaces allowing the user to play with each of the variables described above either independently (open loop), or with computer assistance (closed-loop).
- Created a LabVIEW library for the open-loop and closed-loop control of DC and AC motors, along with DC Generators, for easy programming of future VIs on the project.

Maxwell 3D Simulation (*WKU*) *Fall 2012*

- Collaborated on a Maxwell 3D project to create an animated simulation of the magnetic field lines around an ideal 3-phase induction motor under various loads.
- Both no load and full load simulations generated.

Digital Clock Design (*WKU*) *Fall 2011*

- Design a 2 digit stopwatch style digital clock using Eagle and MultiSIM.
- Print the circuit board using the circuit printing facility and build the board.
- Extra Credit: Design a 4 digit minute/seconds stopwatch instead.

Robot Competitions (*WKU, Georgetown College*) *Fall 2010*

- Taught Mindstorm robotics for over seven years implementing advanced programming challenges such as maze solving algorithms, as well as teaching basic mechanical and electrical concepts.
- Designed a robot using a Basic Stamp and Light Sensors capable of quickly following a line through an obstacle course. Created the body of the robot using machining facility.
- First Place in Freshman speed competition.

PERSONAL PROJECTS

3DBook – 3D Printed Braille Literature Winter 2015

- Open-source software written in OpenSCAD and Python used for the generation of 3D Printer (.stl) files containing full text documents written in braille, printed and bound.
- Enables blind individuals worldwide to access any literature with readily available tools.
- Suggested for admission into the Touch of Genius award by National Braille Press.

OpenScan - Mobile Document Scanner Winter 2015

- Developed collaborative international open-source C++ and Android Application.
- Scans pages and QR codes via video / camera interface quickly and accurately using a variety of methods. Capable of scanning numerous documents with ease in a single use.
- Gives user choice over all settings and access to debugging views. Soon to implement AI to detect proper values for ease of use.

Symbolic Circuit Simulator *Spring 2012*

- Uses Python and Sympy to perform s-Domain circuit analysis
- Useful for deriving the laplacian transformation between input and output of filters and transformer circuits symbolically.

Planetary Orbit Simulation *Spring 2013*

- Uses MatLab and Newton’s law of universal gravitation to plot the motion of any number of gravitationally effected bodies.
- Uses parallel processing and MatLab commands to calculate, export, and plot position data.

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